Nolan Black, Ph.D. | Simulation Engineer

Redacted – U.S.A.

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A research engineer with a strong record in computational engineering, design optimization, and ML for physical simulation. Experienced in HPC programming for numerical analysis and python programming for ML and surrogate development.

Skills & Interests

Computational
- C++ (MPI, PETSc, mfem)
- Python (PyTorch, JAX, SciPy, NumPy)
- MATLAB

CAE Tools
- CFD: Ansys Fluent, OpenFOAM, StarCCM+
- Solid Mechanics: mfem, Ansys Mechanical
- CAD: Solidworks, Gmsh, CREO, Paraview

Professional & Research Experience

Pasteur Labs | Scientific ML Startup — (Remote) Philadelphia, PA
Simulation R&D Engineer — *January 2025 – Present*
- Developed a suite of products for automated simulation, physics informed surrogates, and design optimization across CFD, structural, thermal, and multiphysics domains for aerospace, automotive, and defense industries
- Supported customer deployments and collaborations in which production engineering workflows were adapted for automation, data analysis, and surrogate development
- Designed differentiable software to perform structural optimization of pre-trained ML surrogates in aerospace applications (Python/PyTorch)
- Developed and executed simulation workflows (Ansys suite, Siemens StarCCM+, and OpenFOAM) to generate high-fidelity datasets supporting surrogate modeling for engineering design

Drexel University | GAANN Fellowship — Philadelphia, PA
Graduate Research Assistant — *September 2020 – January 2025*
- Developed and validated high-performance finite-element models for multiscale design optimization (C^{++}/PETSc)
- Investigated novel methods of Machine Learning (ML) for physics-informed simulation (C^{++}/Python/PyTorch)
- Integrated ML techniques to design nature-inspired metamaterials with tailorable physical properties

Lawrence-Livermore Nat. Lab. | Center for Design Optimization — Livermore, CA
NSF PhD INTERN — *January 2024 – May 2024*
- Executed high-performance computing workflows in a U.S. national laboratory environment
- Implemented parallel solvers for mixed finite element methods in a high-performance computing environment (C^{++})
- Developed nonlinear finite element programs for hyperelastic, hierarchical materials for applications in structural optimization

Zimmer Biomet | Zimmer Knee Creations R&D — Exton, PA
Development Engineering Coop — *April 2019 – April 2020*
- Developed functional medical devices using rapid prototyping to perform orthopedic procedures on the hand and wrist
- Coordinated with orthopedic surgeons to established design criteria and verify medical device designs in cadaveric lab testing
- Adapted, tested, and re-validated existing products to meet new standards of quality based on risk management

NAVSEA | Department of the Navy — Philadelphia, PA
Entry-level Engineer — *April 2017 – September 2018*
- Supported US Navy shipboard engineering systems through development and revision of procedures and system documentation
- Coordinated with shipboard engineering personnel to translate operational requirements into clear, auditable engineering documentation

Select Publications

Black, Nolan and Najafi, Ahmad R. (2026), *Length-Scale Effects in Multiscale Structural Optimization Through a Second-Order Homogenization Model*. Int. J. Numer. Methods Eng., 10.1007/s00158-025-04133-5.

Black, Nolan and Najafi, Ahmad R. (2025), *Neural networks for nonlinear homogenization in multiscale structural optimization*. Struct. Multidiscip. Optim., 10.1007/s00158-025-04133-5.

Rehmann, Andrin and **Black**, Nolan and Bjorgaard, Josiah and Angioi, Alessandro and Paleyes, Andrei and Heim, Niklas and Häfner, Dion and Lavin, Alexander (2025), *Surrogate-Based Differentiable Pipeline for Shape Optimization*. arXiv, 10.48550/arXiv.2511.10761.

Black, Nolan and Najafi, Ahmad R. (2023), *Deep neural networks for parameterized homogenization in concurrent multiscale structural optimization*. Struct. Multidiscip. Optim., 10.1007/s00158-022-03471-y.

Black, Nolan and Najafi, Ahmad R. (2022), *Learning finite element convergence with the Multi-fidelity Graph Neural Network*. Comput. Methods Appl. Mech. Eng., 10.1016/j.cma.2022.115120.

Select Conference Presentations

Black, Nolan and Najafi, Ahmad R., "Nonlinear multiscale structural optimization using neural network surrogate models". U.S. National congress on computational mechanics (USNCCM18), Chicago, IL, July 20-24, 2025.

Black, Nolan and Najafi, Ahmad R., "Multiscale structural optimization with strain gradient effects using second order homogenization ". World Congress on Computational Mechanics (WCCM2024), Vancouver, BC, July 21-26, 2024.

Black, Nolan and Najafi, Ahmad R., "Second-order Homogenization for Structural Optimization Applications". Engineering Mechanics Institute Conference 2024 (EMI2024), Chicago, IL, May 28-31, 2024.

Black, Nolan and Najafi, Ahmad R., "Multiscale Design Optimization with Local Failure Criteria through Machine Learning". 17th U.S. National Congress on Computational Mechanics (USNCCM17), Albuquerque, NM, July 23-27, 2023.

Black, Nolan and Najafi, Ahmad R., "Neural Network Surrogates for Multiscale Structural Optimization". 2022 International Mechanical Engineering Congress and Exposition (IMECE 2022), Columbus, OH, Oct. 30 – Nov. 02, 2022.

Black, Nolan and Najafi, Ahmad R., "Multi-Fidelity Graph Neural Networks as Surrogate Models for Finite Element Analysis". 19th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM19), Austin, TX, June 19-24, 2022.

Black, Nolan and Najafi, Ahmad R., "Deep Learning in Concurrent Multiscale Structural Optimization". 19th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM19), Austin, TX, June 19-24, 2022.

Black, Nolan and Najafi, Ahmad R., "Incorporating Architected Materials in Multiscale Structural Optimization with Deep Learning". Engineering Mechanics Institute Conference 2022 (EMI2022), Baltimore, MD, May 31-June 3, 2022.

Scholarships, Fellowships, and Honors

- Fellow, U.S. E.D. Graduate Assistance in Areas of National Need (2020-2025)
- NSF Non-academic Research Internship (INTERN) Award (2024)
- Koerner Family Foundation Fellowship (2022)
- Harry L. Brown, Jr. Memorial Fellowship (2021)
- A.J. Drexel Scholarship, Drexel University (2016–2020)

Professional Affiliations and Service

- Member, The American Society of Mechanical Engineers (2016–Present)
- Student Representative, Graduate Studies Committee, Drexel University (2022–2023)
- Entrepreneurial Lead, NSF I-Corp (2022–2023)

Education

Ph.D., Mechanical Engineering / Computer Science
Drexel University, Philadelphia, PA *September 2020 – December 2024*
Thesis: Multiscale Structural Optimization through Second-order Homogenization and Machine Learning

B.S./M.S., Mechanical Engineering
Drexel University, Philadelphia, PA *September 2016 – June 2020*